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Addressing atmospheric absorption in adaptive rectangular decomposition

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ABSTRACT:
This paper focuses on the adaptive rectangular decomposition (ARD) scheme, a wave-based method employed for acoustic simulations. ARD holds promise for diverse applications. In architectural design, it can forecast acoustical parameters, facilitating the creation of spaces with superior sound quality. Moreover, in the domain of acoustic virtual reality, ARD can offer users a more immersive and lifelike acoustic environment. This enhancement proves advantageous for all of these applications, enabling the simulation of larger environments with heightened precision. Despite its notable efficiency, ARD faces a significant drawback: the absence of atmospheric absorption modeling. The principal aim of this study is to rectify this limitation, thereby augmenting the capabilities of the ARD algorithm. https://doi.org/10.1121/10.0030468

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I. INTRODUCTION

Since the early 1990s, numerous architects and building designers have begun utilizing acoustic simulation software to anticipate specific objective acoustical parameters during the room design phase. However, many errors have arisen due to overreliance on acoustical quality indices like clarity and strength, which do not grasp the intricacies of the room acoustic feature. Auralizations were introduced to address challenges, enabling individuals to audibly experience how a designed room sounds before construction through numerical simulations, and facilitating subjective evaluation of acoustic design quality.

The emergence of acoustic virtual reality (AVR) is notable: equipped with a virtual reality headset, users can navigate simulated environments to experience the predicted acoustic ambiance. This enables straightforward assessment of spatial acoustic variations and the effects of changing parameters (e.g., altering wall materials or rearranging furniture), streamlining the design process. For a dynamic scene-compatible auralization algorithm, readers are referred to Chandak *et al*. (2011). Given that acoustic is influenced by source/receiver placement and other geometry or material parameters, AVR necessitates continuous acoustic simulations, which can be computationally demanding. In the literature, we find two primary approaches to acoustics simulation: geometrical and wave-based methods. See, e.g., Bergman (2018) and Savioja *et al*. (1999). Geometrical methods prove highly efficient in simulating room acoustics; however, since their simplified assumptions, they fail to capture numerous acoustic phenomena, especially at low frequencies, despite their minimal computational demands. On the other hand, wave-based methods approximate partial differential equations (PDEs), inherently capturing all wave phenomena, albeit at a higher computational cost that increases with frequency (Pind *et al*., 2018).

The adaptive rectangular decomposition (ARD), introduced by Raghuvanshi *et al*. (2008) and developed successively in Borrel-Jensen *et al*. (2023), Morales *et al*. (2015), Rabisse *et al*. (2019), Raghuvanshi *et al*. (2009), and Raghuvanshi *et al*. (2010), is a wave-based simulation algorithm noted for its accuracy and computational efficiency, which also incorporates support for absorbing boundary conditions via perfectly matched layers (PMLs). Nevertheless, it possesses inherent limitations such as the absence of atmospheric absorption modeling, which is addressed in this study. To explain the fundamentals of ARD, we report the pioneering algorithm presented in Raghuvanshi *et al*. (2008), which is based on the following steps: (i) the domain is voxelized into a uniform grid with spacing dh; (ii) the grid is partitioned into rectangular subdomains; (iii) assuming homogeneous Neumann boundary conditions, modal analysis is performed in each subdomain to retrieve the mode shapes and modal frequencies; and (iv) a transformation is built to move from the modal-time domain to the space-time domain and back. The governing equation considered in each rectangular subdomain is the wave equation, which is discretized in space, employing a finite-difference method, resulting in a system of coupled ordinary differential equations (ODEs) in time. Then, the ODE system is solved in the modal space through the transformation in (iii), obtaining a system of independent ODE in time. These ODEs are solved using a time marching scheme derived through finite differences, and the solution is transformed back to the spatial domain. The solutions obtained in each subdomain are corrected (interface handling step) to

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impose the propagation of sound waves between all of the subdomains sharing an internal interface.

In Raghuvanshi *et al.* (2009), ARD is presented as an improvement of the previously described method. In particular, (a) the ODE system is solved using the time marching scheme presented in Cieslinski (2011), which exactly discretizes it, unlike the previously employed scheme based on finite differences. This new scheme is unconditionally stable and does not introduce numerical dispersion; (b) the transformation between modal-time time domain and space-time domain is replaced by the Discrete Cosine Transform (DCT), which can be performed more efficiently. The benefits of ARD, according to Raghuvanshi *et al.* (2008) and Raghuvanshi *et al.* (2009), are the following:

- It encompasses all wave phenomena, such as diffraction and reflection, in contrast to geometrical methods;
- even as the grids approach the Nyquist limit, its accuracy remains intact. In contrast, many wave-based methods demand grids with at least five points per wavelength to minimize dispersion, resulting in heightened computational and memory efficiency;
- it can be easily coupled with other simulation methods; and
- it can be easily parallelized: all of the subdomains can be updated independently (coarse-grained parallelism), and all of the cells in modal subdomains can be updated independently (fine-grained parallelism). Refer to Mehra *et al.* (2012b) for further details.

We refer to Mehra *et al.* (2012a) for application/validation of the ARD scheme in scenarios involving the scattering of spherical waves by a rigid sphere and edge diffraction from a right-angled rigid wall. Furthermore, ARD is employed to predict acoustic propagation in a test urban (outdoor) environment, and the agreement with measurement data is reported to be better than the results obtained with the finite-difference time-domain (FDTD) method in two dimensions (2D; Mehra *et al.*, 2012a). In Antani *et al.* (2013), the validation of ARD was conducted within a selected architectural setting, encompassing both indoor and outdoor scenarios. The findings revealed a close correspondence between the simulated decay profile generated by ARD and the measured data. Notably, ARD accurately anticipated mitigating flutter echoes when certain walls were tilted. In Morales *et al.* (2015), a novel efficient parallelization or the ARD algorithm is proposed to improve computational efficiency and obtain scalable performance on distributed architectures. Rabisse *et al.* (2019) use the ARD algorithm to simulate sound propagation in a confined environment while accurately taking account of the acoustic scattering and diffraction effects generated by the presence of geometrical relief on the room surfaces. Finally, Borrel-Jensen *et al.* (2023) aim to enhance simulation realism using the ARD algorithm coupled with the spectral element method near the boundaries to handle complex geometries with realistic boundaries.

Despite these important applications, one of the main limitations of ARD is the lack of atmospheric absorption modeling. The attenuation of sound waves in the free medium becomes significant in large rooms, particularly, at high frequencies. With the growing availability of computational resources, wave-based methods, including ARD, are currently employed for the simulation of wave propagation even at high frequencies. Therefore, to achieve realistic room acoustic simulations, emulation of air absorption is essential (Billon *et al.*, 2008; Hamilton, 2021; Kates and Brandewie, 2020; Kuttruff, 2009; Okuzono, 2023). In this work, we introduce frequency dependent damping to model viscous dissipation in the air. By following Cieslinski (2011), we extend, generalize, and analyze the ARD algorithm for this case. In parallel (and mostly prior) to the development of ARD, different researchers established a similar idea to avoid computing spatial derivatives using finite differences, leading to the pseudo-spectral time-domain (PSTD) method (Hornikx *et al.*, 2016; Hornikx *et al.*, 2010). Unlike ARD, PSTD employs the fast Fourier transform (FFT). This approach has clear advantages over ARD: it does not require imposing even symmetry at the data boundaries, allowing for any type of boundary conditions. Consequently, the coupling between adjacent subdomains does not require an interface handling step to maintain continuity in the pressure field, which can introduce spurious reflections. However, ARD outperforms PSTD in terms of efficiency, and it is possible to derive exact time marching schemes (to update the modal coefficients) because of the restrictive assumptions about the subdomains' boundary conditions. The remainder of the paper is organized as follows. In Sec. II, we introduce the mathematical model and derive the ARD method by introducing the necessary methodology, i.e., the FDTD, the Fourier, and the rectangular domain decomposition (RDD) methods. Notably, the development of the Fourier method incorporates an explicit treatment of atmospheric absorption, which enhances simulation accuracy where absorption effects are significant. We also briefly discuss their stability properties. Next, we generalize the RDD to domains of arbitrary shape with partial absorbing boundaries. In Sec. III, we verify the scheme against manufactured solutions and then, in Sec. IV, we apply the scheme to a case of engineering interest: room acoustics.

II. MODEL PROBLEM AND NUMERICAL DISCRETIZATION

In this section, we introduce the model problem and present the discretization technique that will be employed extensively in this work.

A. Acoustic problem formulation

The propagation of sound waves in an isentropic fluid within a three-dimensional domain, $\Omega = [0, \ell_x] \times [0, \ell_y] \times [0, \ell_z]$, is characterized by the (viscous) *acoustic wave equation*:

$$\begin{cases} \dfrac{\partial^2 p}{\partial t^2}(\underline{x},t) + 2\alpha \dfrac{\partial p}{\partial t}(\underline{x},t) - c^2 \Delta p(\underline{x},t) = f(\underline{x},t), \\ \qquad\qquad\qquad\qquad\qquad t \in \mathbb{R}^+, \ \underline{x} \in \Omega, \\ \dfrac{\partial p}{\partial \underline{x}}(\underline{x},t) = 0, \qquad\qquad t \in \mathbb{R}^+, \ \underline{x} \in \partial\Omega, \\ \left(p, \dfrac{\partial p}{\partial t}\right)(\underline{x},0) = (p_0, v_0)(\underline{x}), \quad \underline{x} \in \Omega, \end{cases} \tag{1}$$

where $p(\underline{x},t)$ denotes the pressure at point $\underline{x} = (x,y,z)$ at time t, c is the *propagation speed*, $f(\underline{x},t)$ denotes an external force, and $\alpha \geq 0$ is the *absorption coefficient*. In Eq. (1), p_0 and v_0 are given functions representing arbitrary initial conditions. The homogeneous Neumann boundary condition models a sound hard boundary. By introducing the pressure velocity, v, we can reformulate Eq. (1) as

$$\begin{cases} \dfrac{\partial v}{\partial t}(\underline{x},t) + 2\alpha v(\underline{x},t) - c^2 \Delta p(\underline{x},t) = f(\underline{x},t), \\ \qquad\qquad\qquad\qquad\qquad t \in \mathbb{R}^+, \quad \underline{x} \in \Omega, \\ \dfrac{\partial p}{\partial t}(\underline{x},t) = v(\underline{x},t), \qquad t \in \mathbb{R}^+, \ \underline{x} \in \Omega, \\ \dfrac{\partial p(\underline{x},t)}{\partial \underline{n}} = 0, \qquad\qquad t \in \mathbb{R}^+, \ \underline{x} \in \partial\Omega, \\ (p,v)(\underline{x},0) = (p_0,v_0)(\underline{x}), \quad \underline{x} \in \Omega. \end{cases} \tag{2}$$

B. FDTD method

In this section, we briefly derive the FDTD method. Although this method is not used directly in simulations, it is essential for developing a simple interface handling technique that will be used in conjunction with the Fourier method.

To find an approximated solution to problem (2) we define a space-time grid (staggered in space) with constant space step size dh and constant time step dt such that

$$\begin{aligned} x_i &= \left(i + \frac{1}{2}\right) \cdot dh, \quad & i \in D_x &= \{0, \ldots, L_x - 1\}, \\ y_j &= \left(j + \frac{1}{2}\right) \cdot dh, \quad & j \in D_y &= \{0, \ldots, L_y - 1\}, \\ z_k &= \left(k + \frac{1}{2}\right) \cdot dh, \quad & k \in D_z &= \{0, \ldots, L_z - 1\}, \\ t^n &= n \cdot dt, & n &\in \mathbb{N}, \end{aligned}$$

where $L_w = \ell_w/dh - 1$ for $w = \{x,y,z\}$. We store the discretized functions in vectors with a row-major order: $D = \{m = i + jL_x + kL_xL_y | i \in D_x, j \in D_y, k \in D_z\} = \{0, \ldots, L-1\}$, where $L = L_xL_yL_z$. Next, for any $m \in D$ and any $n \in \mathbb{N}$, we introduce the notation

$$\varphi_m^n = \varphi_{i,j,k}^n = \varphi(x_i, y_j, z_k, t^n),$$

and define the vector $\underline{\varphi}_{:,j,k}^n$ to denote the restriction of φ to the line parallel to the x axis and passing through the point (x_0, y_j, z_k) for $\varphi \in \{p, v, f\}$. In the same way, we define $\underline{\varphi}_{i,:,k}^n$ and $\underline{\varphi}_{i,j,:}^n$. To reduce dispersion errors in the finite-difference scheme, as suggested in Raghuvanshi *et al.* (2009), we approximate Δp in Eq. (2) using the points centered stencil,

$$(\Delta p)_{i,j,k}^n = \frac{1}{dh^2} \sum_{\ell=-3}^{3} \left(a_h p_{i+\ell,j,k}^n + a_h p_{i,j+\ell,k}^n + a_h p_{i,j,k+\ell}^n \right),$$

where the coefficients are given by $a_{-3} = a_3 = 1/90$, $a_{-2} = a_2 = -3/20$, $a_{-1} = a_1 = 3/2$, and $a_0 = -49/18$. To impose Neumann boundary conditions, we modify the scheme by imposing even symmetry in a standard manner (LeVeque, 2007). Then, to integrate in time equations in Eq. (2), we employ the explicit scheme

$$\begin{cases} p_m^{n+1} = dt v_m^n + p_m^n, & n \in \mathbb{N}, m \in D, \\ v_m^{n+1} = \dfrac{v_m^n + c^2 dt (\Delta p)_m^{n+1} + dt f_m^{n+1}}{1 + 2dt\alpha}, & n \in \mathbb{N}, m \in D, \\ (\underline{p}^0, \underline{v}^0) = (\underline{p}_0, \underline{v}_0), \end{cases} \tag{3}$$

where $\underline{p}_0$ and $\underline{v}_0$ are vectors storing the initial pressure and pressure velocity conditions at the grid points, respectively.

Proposition 1: [Stability of the FDTD scheme (3)] *Scheme (3) is stable provided that the following Courant-Friedrichs-Lewy (CFL) condition is satisfied:*

$$c \frac{dt}{dh} \leq \frac{\sqrt{255}}{34} \approx 0.47. \tag{4}$$

Proof. We start by rewriting the first two equations in Eq. (3):

$$\begin{bmatrix} \underline{v} \\ \underline{p} \end{bmatrix}^{n+1} = [S_{FD}]_{2L} \begin{bmatrix} \underline{v} \\ \underline{p} \end{bmatrix}^{n} + \begin{bmatrix} dt \underline{f}^{n+1} \\ \underline{0} \end{bmatrix}, \tag{5}$$

where $[S_{FD}]_{2L} \in \mathbb{R}^{2L \times 2L}$ is defined as

$$[S_{FD}]_{2L} = \begin{bmatrix} \dfrac{[I]_L + c^2 \dfrac{dt^2}{dh^2}[K]_L}{1 + 2dt\alpha} & \dfrac{c^2 \dfrac{dt}{dh^2}[K]_L}{1 + 2dt\alpha} \\ dt[I]_L & [I]_L \end{bmatrix},$$

where $[I]_L$ is the identity matrix in $\mathbb{R}^{L \times L}$, and $[K]_L \in \mathbb{R}^{L \times L}$ is defined as

$$\begin{aligned} [K]_L = {} & [I]_{L_z} \otimes [I]_{L_y} \otimes [K]_{L_x} + [I]_{L_z} \otimes [K]_{L_y} \otimes [I]_{L_x} \\ & + [K]_{L_z} \otimes [I]_{L_y} \otimes [I]_{L_x}, \end{aligned}$$

where $\otimes$ is the Kronecker product. The matrices $[I]_{L_w}$ for $w = \{x,y,z\}$ are identity matrices of dimension $L_w \times L_w$ while $[K]_{L_w}$ for $w = \{x,y,z\}$ are heptadiagonal matrices having sizes $L_w \times L_w$ and are defined as

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$$[K]_{L_w} = \begin{bmatrix} a_0+a_1 & a_1+a_2 & a_2+a_3 & a_3 & 0 & 0 & 0 & \cdots \\ a_1+a_2 & a_0+a_3 & a_1 & a_2 & a_3 & 0 & 0 & \cdots \\ a_2+a_3 & a_1 & a_0 & a_1 & a_2 & a_3 & 0 & \cdots \\ a_3 & a_2 & a_1 & a_0 & a_1 & a_2 & a_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

To prove the stability of Eq. (5), we compute the spectral radius $\sigma_{S_{FD}}$ of $[S_{FD}]_{2L}$ and check if it is less than or equal to one, i.e., $\sigma_{S_{FD}} \leq 1$. Let us start by considering the case where $\alpha = 0$ and observing that the eigenvalues $\lambda_{S_{FD}}$ of $[S_{FD}]_{2L}$ are given by

$$\lambda_{S_{FD}}^2 - \left(2 + \lambda_K \left(c\frac{dt}{dh}\right)^2\right)\lambda_{S_{FD}} + 1 = 0,$$

where λ_K is an eigenvalue of $[K]_L$. It can be revealed that $|\lambda_{S_{FD}}| = 1$ if the discriminant is nonpositive, i.e.,

$$\left(\frac{1}{2}\lambda_K \left(c\frac{dt}{dh}\right)^2\right) + \lambda_K \left(c\frac{dt}{dh}\right)^2 \leq 0.$$

This condition is satisfied if and only if $|\lambda_K| \leq 4/(cdt/dh)^2$ for any λ_K. This is equivalent to requiring that the spectral radius σ_K of $[K]_L$ is $\sigma_K \leq 4/(cdt/dh)^2$, that is, $cdt/dh \leq 2/\sqrt{\sigma_K}$. From the Gershgorin's theorem (Bell, 1965), it follows that

$$\sigma_K = 3\sum_{i=-3}^{3} |a_i| = \frac{272}{15},$$

hence, $cdt/dh \leq \sqrt{255}/34$. The proof for the case $\alpha > 0$ is analogous (despite being longer) and omitted here for brevity: it can be shown that Eq. (4) is a lower bound for a less restrictive CFL condition. □

C. Fourier method

In this section, based on the work by Raghuvanshi *et al.* (2008), Raghuvanshi *et al.* (2009), and Savioja *et al.* (2010), we modify the classical Fourier method to support air absorption. We start by introducing the following mode shapes, cf. Asmar (2005):

$$q_{\mu\nu\xi}(\underline{x}) = \sqrt{\frac{8}{\ell_x\ell_y\ell_z}}\varepsilon_\mu\varepsilon_\nu\varepsilon_\xi \cos\left(\frac{\mu\pi}{\ell_x}x\right)\cos\left(\frac{\nu\pi}{\ell_y}y\right) \times \cos\left(\frac{\xi\pi}{\ell_z}z\right), \tag{6}$$

where $\varepsilon_0 = 1/\sqrt{2}$, $\varepsilon_\mu = 1$ for $\mu > 0$, and express $p(\underline{x}, t)$ as a linear combination of the mode shapes

$$p(\underline{x}, t) = \sum_{\xi=0}^{\infty}\sum_{\nu=0}^{\infty}\sum_{\mu=0}^{\infty} P_{\mu\nu\xi}(t)q_{\mu\nu\xi}(\underline{x}), \tag{7}$$

where the modal coefficients $P_{\mu\nu\xi}(t)$ can be computed as

$$P_{\mu\nu\xi}(t) = \int_0^{\ell_z}\int_0^{\ell_y}\int_0^{\ell_x} p(\underline{x}, t)q_{\mu\nu\xi}(\underline{x})\, dx\, dy\, dz. \tag{8}$$

Next, we do the same for f and v to obtain $F_{\mu\nu\xi}$ and $V_{\mu\nu\xi}$. Then, we reformulate problem (1) for $t \in \mathbb{R}^+$, $\mu,\nu,\ \xi \in \mathbb{N}^+$, as

$$\begin{cases} \ddot{P}_{\mu\nu\xi}(t) + 2\alpha\dot{P}_{\mu\nu\xi}(t) + \omega_{0;\mu\nu\xi}^2 P_{\mu\nu\xi}(t) = F_{\mu\nu\xi}(t), \\ (P_{\mu\nu\xi}, \dot{P}_{\mu\nu\xi})(0) = (P_{\mu\nu\xi,0}, V_{\mu\nu\xi,0}), \end{cases} \tag{9}$$

where $P_{\mu\nu\xi,0}, V_{\mu\nu\xi,0}$ are given coefficients representing arbitrary initial conditions, and $\omega_{0;\mu\nu\xi} = c\pi\sqrt{\mu^2/\ell_x^2 + \nu^2/\ell_y^2 + \xi^2/\ell_z^2}$. We notice that for each tuple (μ, ν, ξ), Eq. (9) represents the equation of motion for a single degree-of-freedom system (forced damped harmonic oscillator). Employing the staggered space grid that is defined in Sec. II B, we rewrite the mode shapes in Eq. (6) as

$$q_{\mu\nu\xi,ijk} = \sqrt{\frac{8}{L_xL_yL_z}}\varepsilon_\mu\varepsilon_\nu\varepsilon_\xi \cos\left(\frac{\pi\mu}{L_x}\left(i + \frac{1}{2}\right)\right) \times \cos\left(\frac{\pi\nu}{L_y}\left(j + \frac{1}{2}\right)\right)\cos\left(\frac{\pi\xi}{L_z}\left(k + \frac{1}{2}\right)\right).$$

Notice that because we are dealing with discrete signals in space, the spatial frequencies are limited by the Nyquist frequency. For example, considering the x axis, we have that $\pi\mu/L_x < \pi \Rightarrow \mu < L_x$. Hence, $\mu \in D_x$, $\nu \in D_y$, and $\xi \in D_z$. Recalling Eq. (7), we express $p_m(t)$ as

$$p_m(t) = \sum_{\eta=0}^{L-1} q_{\eta,m}P_\eta(t), \tag{10}$$

where P_η are the modal coefficients of the pressure p. Conversely, recalling Eq. (8), the modal coefficients $P_\eta(t)$ can be computed through

$$P_\eta(t) = \sum_{m=0}^{L-1} q_{\eta,m}p_m(t). \tag{11}$$

The same can be performed for the force f and pressure velocity v. Notice that Eq. (11) corresponds to the DCT in its DCT-II form, whereas Eq. (10) corresponds to the inverse DCT in its DCT-III form (Bull and Zhang, 2021). To conclude, we reformulate problem (9) as

$$\begin{cases} \ddot{P}_\eta(t) + 2\alpha\dot{P}_\eta(t) + \omega_{0;\eta}^2 P_\eta(t) = F_\eta(t), \\ \quad t \in \mathbb{R}^+,\ \eta \in D, \\ (\underline{P}, \underline{\dot{P}})(0) = (\underline{P}_0, \underline{V}_0), \end{cases} \tag{12}$$

or in its first-order variant,

$$\begin{cases} \begin{bmatrix} \dot{V}_\eta(t) \\ \dot{P}_\eta(t) \end{bmatrix} = \begin{bmatrix} -2\alpha & -\omega_{0;\eta}^2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} V_\eta(t) \\ P_\eta(t) \end{bmatrix} + \begin{bmatrix} F_\eta(t) \\ 0 \end{bmatrix}, \\ \qquad t \in \mathbb{R}^+, \eta \in D, \\ (\underline{P}(0), \underline{V}(0)) = (\underline{P}_0, \underline{V}_0). \end{cases} \tag{13}$$

Following the idea presented in Cieslinski (2011), to integrate in time system (13), we consider the general scheme

$$\begin{bmatrix} V_\eta \\ P_\eta \end{bmatrix}^{n+1} = [S_\eta] \begin{bmatrix} V_\eta \\ P_\eta \end{bmatrix}^{n} + [T_\eta] F_\eta^n \quad \text{for } n \geq 0. \tag{14}$$

The definitions of the matrices $[S_\eta]$ and $[T_\eta]$ depend on the four cases:

(i) $\eta = 0$ and $\alpha = 0$,

$$[S_0] = \begin{bmatrix} 1 & 0 \\ dt & 1 \end{bmatrix}, \quad [T_0] = \begin{bmatrix} dt \\ \dfrac{dt^2}{2} \end{bmatrix};$$

(ii) $\eta = 0$ and $\alpha > 0$,

$$[S_0] = \begin{bmatrix} e^{-2\alpha dt} & 0 \\ \dfrac{1 - e^{-2\alpha dt}}{2\alpha} & 1 \end{bmatrix},$$

$$[T_0] = \begin{bmatrix} \dfrac{1 - e^{-2\alpha dt}}{2\alpha} \\ \dfrac{e^{-2\alpha dt} - 1}{4\alpha^2} + \dfrac{1}{2\alpha} dt \end{bmatrix};$$

(iii) $\eta > 0$ and $0 \leq \alpha < \omega_{0;\eta}$,

$$[S_\eta] = e^{-\alpha dt}\cos(\omega_\eta dt) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + e^{-\alpha dt}\sin(\omega_\eta dt) \begin{bmatrix} -\dfrac{\alpha}{\omega_\eta} & -\left(\omega_\eta + \dfrac{\alpha^2}{\omega_\eta}\right) \\ \dfrac{1}{\omega_\eta} & \dfrac{\alpha}{\omega_\eta} \end{bmatrix}, \quad [T_\eta] = -\frac{1}{\omega_{0;\eta}^2} \begin{bmatrix} e^{-\alpha dt}\left(\omega_\eta + \dfrac{\alpha^2}{\omega_\eta}\right)\sin(\omega_\eta dt) \\ 1 - e^{-\alpha dt}\left(\cos(\omega_\eta dt) + \dfrac{\alpha}{\omega_\eta}\sin(\omega_\eta dt)\right) \end{bmatrix},$$

where $\omega_\eta = \sqrt{\omega_{0;\eta}^2 - \alpha^2}$ are the damped frequencies;

(iv) $\eta > 0$ and $\alpha > \omega_{0;\eta}$,

$$[S_\eta] = \frac{1}{2} \begin{bmatrix} \left(e^{-\alpha_{1;\eta}dt} + e^{-\alpha_{2;\eta}dt}\right) & -\left(\alpha_{1;\eta}e^{-\alpha_{1;\eta}dt} + \alpha_{2;\eta}e^{-\alpha_{2;\eta}dt}\right) \\ 0 & 2\left(e^{-\alpha_{1;\eta}dt} + e^{-\alpha_{2;\eta}dt}\right) \end{bmatrix} + \frac{1}{2\alpha_{d;\eta}} \begin{bmatrix} \alpha\left(e^{-\alpha_{1;\eta}dt} - e^{-\alpha_{2;\eta}dt}\right) & -\alpha\left(\alpha_{2;\eta}e^{-\alpha_{2;\eta}dt} - \alpha_{1;\eta}e^{-\alpha_{1;\eta}dt}\right) \\ \left(e^{-\alpha_{2;\eta}dt} - e^{-\alpha_{1;\eta}dt}\right) & \left(\alpha_{2;\eta}e^{-\alpha_{2;\eta}dt} - \alpha_{1;\eta}e^{-\alpha_{1;\eta}dt}\right) \end{bmatrix},$$

$$[T_\eta] = \frac{1}{\omega_{0;\eta}^2} \begin{bmatrix} \dfrac{1}{2}\left(\alpha_{1;\eta}e^{-\alpha_{1;\eta}dt} + \alpha_{2;\eta}e^{-\alpha_{2;\eta}dt} + \dfrac{\alpha}{\alpha_{d;\eta}}\left(\alpha_{2;\eta}e^{-\alpha_{2;\eta}dt} - \alpha_{1;\eta}e^{-\alpha_{1;\eta}dt}\right)\right) \\ 1 - e^{-\alpha_{1;\eta}dt} - e^{-\alpha_{2;\eta}dt} - \dfrac{1}{2\alpha_{d;\eta}}\left(\alpha_{2;\eta}e^{-\alpha_{2;\eta}dt} - \alpha_{1;\eta}e^{-\alpha_{1;\eta}dt}\right) \end{bmatrix},$$

where and $\alpha_{d;\eta} = \sqrt{\alpha^2 - \omega_{0;\eta}^2}$, $\alpha_{1,2;\eta} = \alpha \pm \alpha_{d;\eta}$.

Proposition 2: (Stability of the Fourier method) *Scheme (14) is unconditionally stable.*

Proof. For each of the four cases, we compute the eigenvalues of $[S_\eta]$ and check that their absolute values are less than or equal to one.

(i) It is easy to check that $|\lambda_{1,2}| = 1$;
(ii) similarly, we have $|\lambda_1| = e^{-2\alpha dt} < 1$ and $|\lambda_2| = 1$;
(iii) for $\alpha = 0$, the characteristic equation is

$$|[S_\eta] - \lambda[I]| = \lambda^2 - 2\lambda\cos(\omega_\eta dt) + 1 = 0.$$

Hence, $\lambda_{1,2} = \cos(\omega_\eta dt) \pm \jmath \sin(\omega_\eta dt)$ and then $|\lambda_{1,2}| = 1$. A similar result holds for $0 < \alpha < \omega_{0;\eta}$;
(iv) in this case, it can be proven that

$$\lambda_1 = e^{-dt\left(\alpha + \sqrt{\alpha^2 - \omega_{0;\eta}^2}\right)}, \quad \lambda_2 = e^{-dt\left(\alpha - \sqrt{\alpha^2 - \omega_{0;\eta}^2}\right)},$$

such that $|\lambda_{1,2}| < 1$ for each value of $\alpha > \omega_{0;\eta}$. □

D. RDD

The Fourier method presented in Sec. II C is not suited for complex geometries: indeed, it is valid only for parallelepipedal domains. To overcome this limitation. the RDD has been introduced and employed in the literature (Raghuvanshi *et al.*, 2008; Raghuvanshi *et al.*, 2009; Savioja *et al.*, 2010). Here, for the sake of completeness, we briefly review the main aspects of RDD by considering the case in which Ω is a parallelepiped of dimension $2\ell_x$, ℓ_y, and ℓ_z as represented in Fig. 1, and adapting it to handle the discretization in Sec. II B.

1. Derivation of RDD for the FDTD scheme and extension to the Fourier method

We consider a staggered uniform spatial grid of step size dh along all directions as follows:

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$$x_i = \left(i + \frac{1}{2}\right)dh, \quad i \in D_x = \{0, \ldots, 2L_x - 1\},$$
$$y_j = \left(j + \frac{1}{2}\right)dh, \quad j \in D_y = \{0, \ldots, L_y - 1\},$$
$$z_k = \left(k + \frac{1}{2}\right)dh, \quad k \in D_z = \{0, \ldots, L_z - 1\}.$$

We store the grid functions in vectors with a row-major order,

$$\begin{aligned} D &= \left\{m = i + j2L_x + k2L_xL_y | i \in D_x, j \in D_y, k \in D_z\right\} \\ &= \{0, \ldots, 2L - 1\}, \end{aligned}$$

where $L = L_xL_yL_z$, and the first-order FDTD scheme (3) is considered. For the sake of presentation, we partition the domain Ω into two distinct nonoverlapping subdomains: Ω_1, corresponding to $x \leq \ell_x$, and Ω_2, corresponding to $x \geq \ell_x$. These subdomains have dimensions ℓ_x, ℓ_y, and ℓ_z as illustrated in Fig. 1, and they share the interface Γ, which is a surface parallel to the yz plane. We use the subscript d to denote a quantity in the subdomain Ω_d, $d = 1, 2$, and employ the notation

$$\varphi^n_{i,j,k;d} = \varphi(x_i, y_j, z_k, t^n) \quad \text{for } (x_i, y_j, z_k) \in \Omega_d,$$

where $\varphi = \{p, v, f\}$. We proceed by decoupling the matrix $[K]_{2L_x}$ of dimensions $2L_x \times 2L_x$ into a block-diagonal form matrix $[A]_{2L_x}$ and a residual matrix $[C]_{2L_x}$ such that

$$[K]_{2L_x} = [A]_{2L_x} + [C]_{2L_x},$$

where

$$[A]_{2L_x} = \begin{bmatrix} [K]_{L_x} & [0]_{L_x} \\ [0]_{L_x} & [K]_{L_x} \end{bmatrix},$$

and $[C]_{2L_x}$ is computed by difference. We can interpret the matrix $[A]_{2L_x}$ as imposing the homogeneous Neumann boundary conditions (even symmetry) at the interface Γ, whereas the matrix $[C]_{2L_x}$ can be interpreted as imposing the transmission conditions. We introduce the *residual* $r^{n+1}_{i,j,k}$, defined as

$$\underline{r}^{n+1}_{:,j,k} = \left(\frac{c}{dh}\right)^2 [C]_{2L_x}\, \underline{p}^{n+1}_{:,j,k}, \quad n \in \mathbb{N}, \qquad j \in D_y,\ k \in D_z. \tag{15}$$

Notice that the residual cannot be computed independently in each subdomain as it depends also on the pressure in the neighboring region. However, as the matrix $[C]_{2L_x}$ is sparse, only six pressure values are considered at positions $L_x - 3$, $L_x - 2$, and $L_x - 1$ for Ω_1 and at positions 0, 1, and 2 for Ω_2. With this formalism, the FDTD scheme (3) can be rewritten as

$$\begin{cases} p^{n+1}_{i,j,k;d} = dt v^n_{i,j,k;d} + p^n_{i,j,k;d}, \\ \hat{v}^{n+1}_{i,j,k;d} = \dfrac{v^n_{i,j,k;d} + c^2 dt (\Delta p)^{n+1}_{i,j,k;d} + dt f^{n+1}_{i,j,k;d}}{1 + 2dt\alpha}, \\ (\underline{p}^0_d, \underline{v}^0_d) = (\underline{p}_{d;0}, \underline{v}_{d;0}). \end{cases} \tag{16}$$

For $n \in \mathbb{N}$, $i \in D_d$, $j \in D_y$, $k \in D_z$, and $d = 1{,}2$. We then use the residual (15) to correct the solution given by Eq. (16) near the interfaces,

$$v^{n+1}_{i,j,k} \leftarrow \hat{v}^{n+1}_{i,j,k} + \frac{dt}{1 + 2dt\alpha} r^{n+1}_{i,j,k}. \tag{16a}$$

We refer to the last step as *interface correction step*. This technique can be combined with the Fourier method introduced in Sec. II C. Indeed, one can apply the scheme (14) to compute the solution in each subdomain Ω_d, $d = 1{,}2$, and use the residual (16a) to update it near the interfaces. In this case, the coupling is not exact as the residual is built on the FDTD method. This introduces erroneous (spurious) reflections at the interface between subdomains, grid dispersion, and a restriction on the simulation time step given by Eq. (4). The latter result comes from Mehra *et al.* (2012a) and Mehra *et al.* (2012b), which suggest that the CFL condition of the ARD corresponds to that of the FDTD scheme employed to derive the residual.

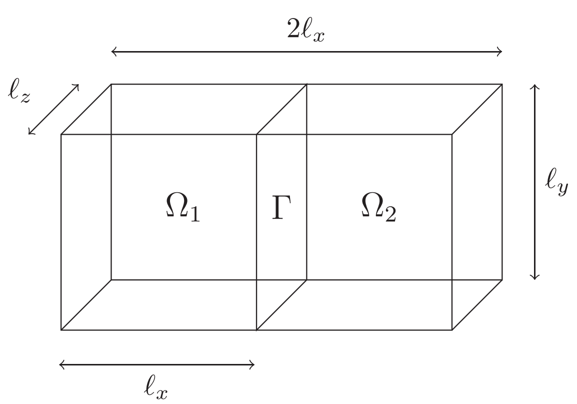

FIG. 1. Representation of the domain Ω partitioned into two nonoverlapping subdomains Ω_1, Ω_2, sharing the interface Γ.

2. Generalization of RDD for domains of arbitrary shape with partial absorbing boundaries

In this section, we explain how to generalize the scheme introduced previously for the case of arbitrarily shaped domains with partial absorbing boundaries. The model introduces the latter through PML conditions. For presentation's sake, we detail the implementation of PML conditions in the Appendix.

We acknowledge that using PML conditions is an approximation for modeling absorbing boundaries. As the Fourier method operates under the assumption of perfectly reflecting boundary conditions, the only way to simulate boundary absorption is by connecting Fourier subdomains to absorbing subdomains such as PML. Although this approach does not fully capture the complex interactions at real room boundaries, it remains effective for simulating partial absorption.

As proposed in Raghuvanshi *et al.* (2009), partial boundary absorption is treated by modifying the interface correction step,

$$v_{i,j,k}^{n+1} \leftarrow \hat{v}_{i,j,k}^{n+1} + \beta \frac{dt}{1+2dt\alpha} r_{i,j,k}^{n+1},$$

where $0 \leq \beta \leq 1$ is the dimensionless boundary absorption coefficient (e.g., full reflection with $\beta = 0$ or full absorption with $\beta = 1$). This scheme, called ARD, consists of two primary stages:

- *Preprocessing* (Fig. 4)
  (1) The input scene is voxelized into grid cells at grid resolution dh;
  (2) the grid cells are grouped into N_d rectangles, corresponding to air subdomains Ω_d for $d = 1, \ldots, N_d$, and PML subdomains Ω_{PML} are generated for each boundary; for each Ω_d, we store the grid functions and other variables needed to run the solution scheme; and
  (3) the interfaces between adjacent subdomains are created by examining adjacency relationships;
  (4) *simulation loop*;
  (5) for each subdomain Ω_d, we update the solution by employing the Fourier method in Sec. II C, and the FDTD scheme described in Sec. II A for PML subdomains; the chosen time step dt must satisfy Eq. (4); and
  (6) for each interface, we perform the interface correction step.

III. VALIDATION

In this section, we perform numerical experiments to validate the efficiency of the proposed scheme (14), either with or without the interface correction step (16a), in terms of convergence, accuracy, and efficiency.

A. Fourier method

In this section, we perform some convergence tests for the scheme (14). We consider a domain $\Omega = (0, 10\ \mathrm{m})^3$, with propagation speed $c = 1\ \mathrm{m/s}$, no prescribed external force, and the following standing wave solution associated with mode $(\mu, \nu, \xi) = (7, 2, 2)$:

TABLE I. Test case of Sec. III A, showing computed L^∞ error for the Fourier method [Eq. (14)] by varying dt for $dh = 0.125$ and different values of α.

α \ dt	0.001	0.002	0.005	0.01
0	5.5927×10^{-15}	2.9560×10^{-15}	1.2768×10^{-15}	6.5226×10^{-16}
$0.5\omega_0$	4.3993×10^{-15}	2.2204×10^{-15}	1.2629×10^{-15}	4.9960×10^{-16}
$1.5\omega_0$	5.1070×10^{-15}	1.8874×10^{-15}	1.0825×10^{-15}	4.9960×10^{-16}

TABLE II. Test case of Sec. III A, showing computed L^∞ error for the Fourier method [Eq. (14)] by varying dh for $dt = 0.01$ and different values of α.

α \ dh	0.03125	0.0625	0.125	0.25
0	5.8287×10^{-16}	6.3838×10^{-16}	6.5226×10^{-16}	5.2736×10^{-16}
$0.5\omega_0$	4.9960×10^{-16}	6.2450×10^{-16}	4.9960×10^{-16}	5.6899×10^{-16}
$1.5\omega_0$	4.1633×10^{-16}	4.9960×10^{-16}	4.9960×10^{-16}	3.8858×10^{-16}

$$p(x,y,z,t) = \cos\left(\frac{7\pi}{10}x\right)\cos\left(\frac{2\pi}{10}y\right) \times \cos\left(\frac{2\pi}{10}z\right)e^{-\alpha t}\cos(\omega t), \quad (17)$$

where $\omega = \sqrt{\omega_0^2 - \alpha^2}$ rad/s and $\omega_0 = \sqrt{57}/10c\pi$ rad/s. The L^∞ error at the final time $T = 0.1$ s is computed as $\mathrm{Err}_{L^\infty}(T) = \max_{x_i,y_j,z_k \in \Omega}|p_h(x_i,y_j,z_k,T) - p(x_i,y_j,z_k,T)|$, where p_h represents the numerical solution, p is the analytical solution in Eq. (17). Tables I and II summarize the computed L^∞ error for different choices of dh and dt and different values of the absorption coefficient α. We notice how the error stays on the order of magnitude of the machine precision (2.2204×10^{-16}). This confirms that the Fourier method is exact in space and time. The small numerical error is a consequence of the floating-point arithmetic.

B. Fourier method with RDD

In this section, we first perform a convergence test by combining the schemes (14) and (16a), considering the same solution presented in Sec. III A with $\alpha = 0\ \mathrm{s}^{-1}$. The domain $\Omega = (0, 10\ \mathrm{m})^3$ is partitioned into two subdomains as shown in Fig. 1: $\Omega_1 = (0, 5\ \mathrm{m}) \times (0, 10\ \mathrm{m})^2$ and $\Omega_2 = (5\ \mathrm{m}, 10\ \mathrm{m}) \times (0, 10\ \mathrm{m})^2$. We have chosen a test case that is suitable for assessing interface errors: a spherical wave generated by an omnidirectional source. In this case, the propagation speed is $c = 5\ \mathrm{m/s}$, the prescribed initial velocity is zero, and the initial pressure is

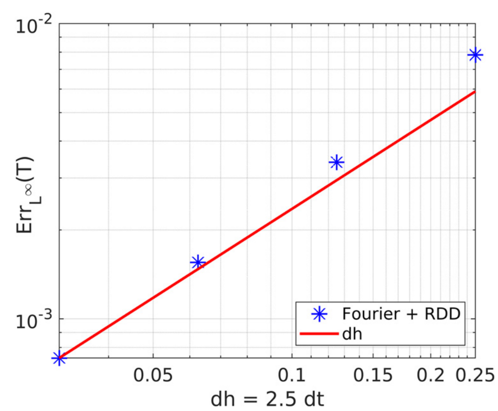

FIG. 2. (Color online) Test case of Sec. III B, showing computed L^∞ error at the final time $T = 1$ s for the Fourier method with RDD by varying $dh = 2.5dt$.

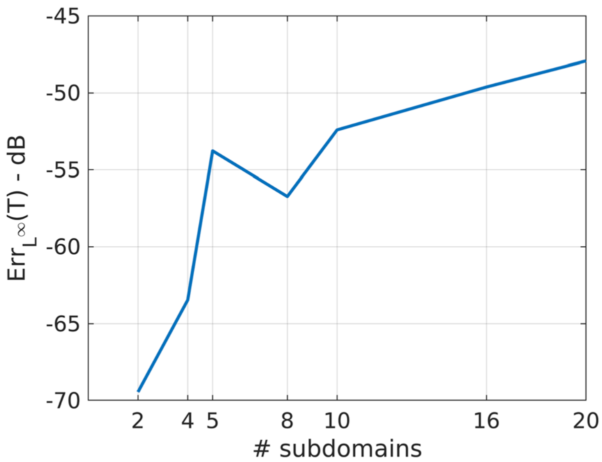

FIG. 3. (Color online) Test case of Sec. III B, showing computed L^∞ error at the final time 1 s for Fourier method with RDD convergence test varying the number of subdomains.

$$p_0(x,y,z)=\frac{1}{(2\pi)^{3/2}\sigma_x\sigma_y\sigma_z}\times\exp\left(-\frac{1}{2}\left[\frac{(x-\mu_x)^2}{\sigma_x^2}+\frac{(y-\mu_y)^2}{\sigma_y^2}+\frac{(z-\mu_z)^2}{\sigma_z^2}\right]\right),$$

where $\mu_x = 2.5$ m, $\mu_y = \mu_z = 0.5$ m and $\sigma_x = \sigma_y = \sigma_z = 0.5$ m. We obtain a spherical wave: before hitting the boundaries, the wavefronts are concentric spheres centered on the source. In Fig. 2, we plot the computed L^∞ error at the final time $T = 1$ s for different choices of dh and dt. We have chosen a longer simulation time to accentuate the interface errors. We conclude that the Fourier method with RDD converges to the ground truth solution with order $\mathcal{O}(dh + dt)$.

Then, in Fig. 3, we show the computed L^∞ error in dB scale $[\mathrm{Err}_{L^\infty}(T)\,\mathrm{dB} = 20\log_{10}(\mathrm{Err}_{L^\infty}(T))]$ by increasing the number of subdomains, fixing $dh = 0.0625$ m and $dt = 5\times10^{-3}$ s. We notice that the error increases slowly with the number of subdomains and can be considered inaudible as a result of the acoustic masking phenomenon: the spurious reflections are masked by the incident waves, where the latter is higher in loudness (Gelfand, 2017).

Furthermore, it is important to note that because spurious reflections and legitimate waves are attenuated in a similar manner, the ratio of energy from spurious reflections to the total acoustic energy is expected to remain constant over time.

IV. APPLICATION TO ROOM ACOUSTICS

A. Introducing frequency dependent air absorption

The current standard in acoustics, which describes the attenuation of sound (ISO, 1993), relates attenuation to the frequency of the sound and the temperature, humidity, and pressure of the air. In particular, the attenuation resulting from classical and rotational absorption is proportional to the square of the frequency. For a more detailed explanation of sound attenuation in air, refer to Evans and Sutherland (1970).

For the purposes of this study, we focus solely on the relationship between air absorption and frequency. The absorption term in Eq. (1) alone cannot accurately model atmospheric absorption as the absorption is constant with respect to frequency. This necessitates a modification of the PDE.

As detailed in Ruiz (1969), to introduce a damping force that increases with ω as ω^{2n}, we need to include the term $(-1)^n(\partial^{2n+1}p/\partial t^{2n+1})$. For a damping force that increases with ω as ω^2, we need to introduce the term $-\partial^3 p/\partial t^3$. However, Bensa *et al.* (2003) found that this term makes the PDE ill-posed and unstable at very high sampling rates. Thus, it should be replaced by a third-order mixed partial derivative: $\partial^3 p/\partial x^2\partial t$, which in our context becomes $\partial\Delta p/\partial t$. This approach is further elaborated in Bensa *et al.* (2005), Desvages and Bilbao (2016), Desvages *et al.* (2016), and Smith (2010). Consequently, our PDE becomes

$$\begin{cases}\dfrac{\partial^2 p}{\partial t^2}(\underline{x},t)+2\alpha_1\dfrac{\partial p}{\partial t}(\underline{x},t)-c^2\Delta p(\underline{x},t)\\ \quad -2c^2\alpha_2\dfrac{\partial\Delta p(\underline{x},t)}{\partial t}=f(\underline{x},t), \quad t\in\mathbb{R}^+,\ \underline{x}\in\Omega,\\ \dfrac{\partial p}{\partial \underline{n}}(\underline{x},t)=0, \qquad t\in\mathbb{R}^+,\ \underline{x}\in\partial\Omega,\\ \left(p,\dfrac{\partial p}{\partial t}\right)(\underline{x},0)=(p_0,v_0)(\underline{x}), \quad \underline{x}\in\Omega,\end{cases} \tag{18}$$

where α_1 and α_2 are parameters controlling the relation between attenuation and frequency. Transforming to the modal domain, we obtain the following ODE:

$$\ddot{P}_{\mu\nu\xi}(t)+2\underbrace{\left(\alpha_1+\alpha_2\omega^2_{0;\mu\nu\xi}\right)}_{\alpha_{\mu\nu\xi}}\dot{P}_{\mu\nu\xi}(t) + \omega^2_{0;\mu\nu\xi}P_{\mu\nu\xi}(t)=F_{\mu\nu\xi}(t). \tag{19}$$

This equation represents a damped harmonic oscillator, allowing us to use the Fourier method in Eq. (14) to obtain its numerical solution. The only difference is that the value of α is mode specific. However, as the FDTD scheme in Eq. (3) does not consider the frequency dependent absorption term introduced in Eq. (18), the interface correction step in Eq. (16a) is not compatible anymore. The derivation of a new interface correction step is out of the scope of the present work. As a workaround, we neglect the new term such that Eq. (16a) becomes

$$v^{n+1}_{i,j,k}\leftarrow \hat{v}^{n+1}_{i,j,k}+\frac{dt}{1+2dt\alpha_1}r^{n+1}_{i,j,k}. \tag{20}$$

B. Numerical results

In this section, we test the previously derived ARD by enhancing the c++ software *ARD-simulator* freely available online[1] to include (frequency dependent) air absorption in the computational domain.

This new version is also available online[2]. We consider the simplified hall Ω depicted in Fig. 4(a) as having a constant height of 10 m, a volume of 7600 m^3, and a surface area of

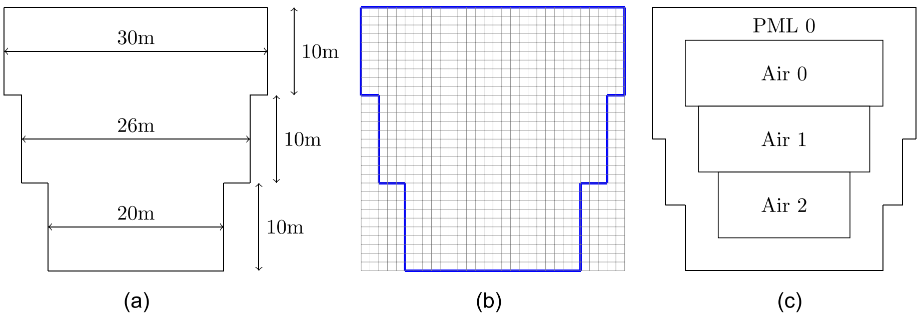

FIG. 4. (Color online) Preprocessing step of ARD in a test scenario (top view). (a) Hall (domain), (b) voxelization with $dh = 1$m, and (c) partitioning are depicted.

760 m^2. We partition Ω into three rectangles, Ω_1, Ω_2, and Ω_3, as depicted in Fig. 4(c). We choose the propagation speed $c = 343.5$ m/s, the grid spacing $dh = 0.2$ m, the time step $dt = 2 \times 10^{-4}$ s, the boundaries absorption coefficient is $\beta = 0$, and $\alpha_1 = 0\ \mathrm{s}^{-1}$. A visual comparison between three cases, where $\alpha_2 = \{1, 2, 5\} \times 10^{-6}\ \mathrm{s}^{-1}$, is shown in Fig. 5. The applied force is of the form $f(\underline{x}, t) = \delta(\underline{x} - \underline{x}_0)e^{-((t-t_0)/\sigma)^2}$, where $\delta(\cdot)$ is the three-dimensional Dirac delta function; $\underline{x}_0$ is the position of the source, i.e., the center of the lower rectangle; $t_0 = 6dh/(cdt)$; and $\sigma = 3dh/(cdt)$. This simulation, irrespective of the chosen physical parameters (speed of propagation, air, and boundaries absorption), takes approximately 50 min to simulate 1 s of the event on a mid-range laptop (parallel computation on eight cores). With this example, we have demonstrated the applicability of this algorithm to realistic cases. We are convinced that its natively parallel structure will make it suitable for (almost) real-time simulations.

To validate the capability of our method in effectively modeling atmospheric absorption, we have compared the experimental energy decay curves (defined as in Kuttruff, 2009) with the theoretical ones ($e^{-2\alpha t}$) in four octave bands (Fig. 6).

V. CONCLUSION

We have proposed an improved version of ARD that adds support for modeling atmospheric absorption, which leads to an increased simulation accuracy at higher frequencies where absorption is significant, while not decreasing the computational efficiency and retaining all of the other benefits (and other limitations) of the original ARD

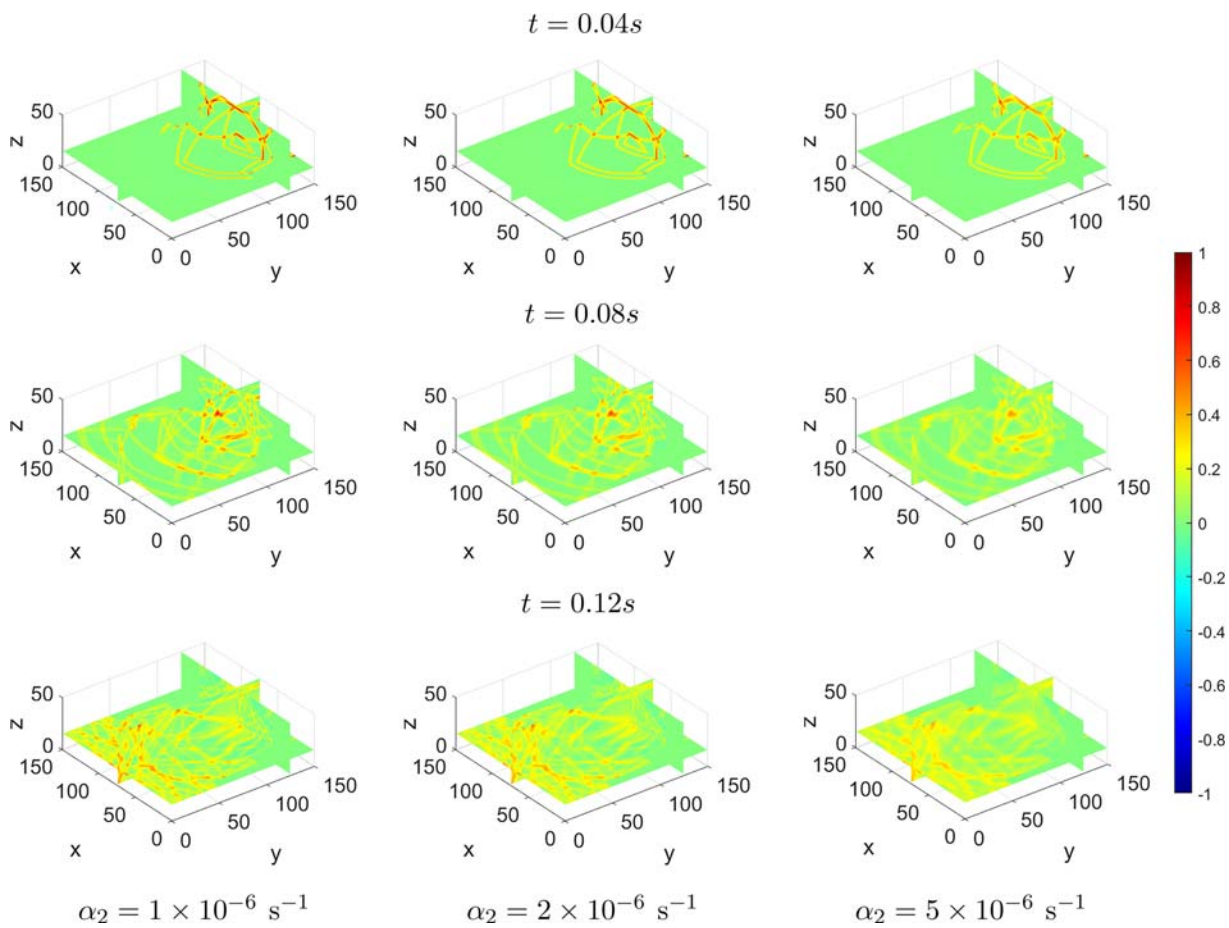

FIG. 5. (Color online) Comparison of simulation with $\alpha_2 = \{1, 2, 5\} \times 10^{-6}\ \mathrm{s}^{-1}$ at different instants.

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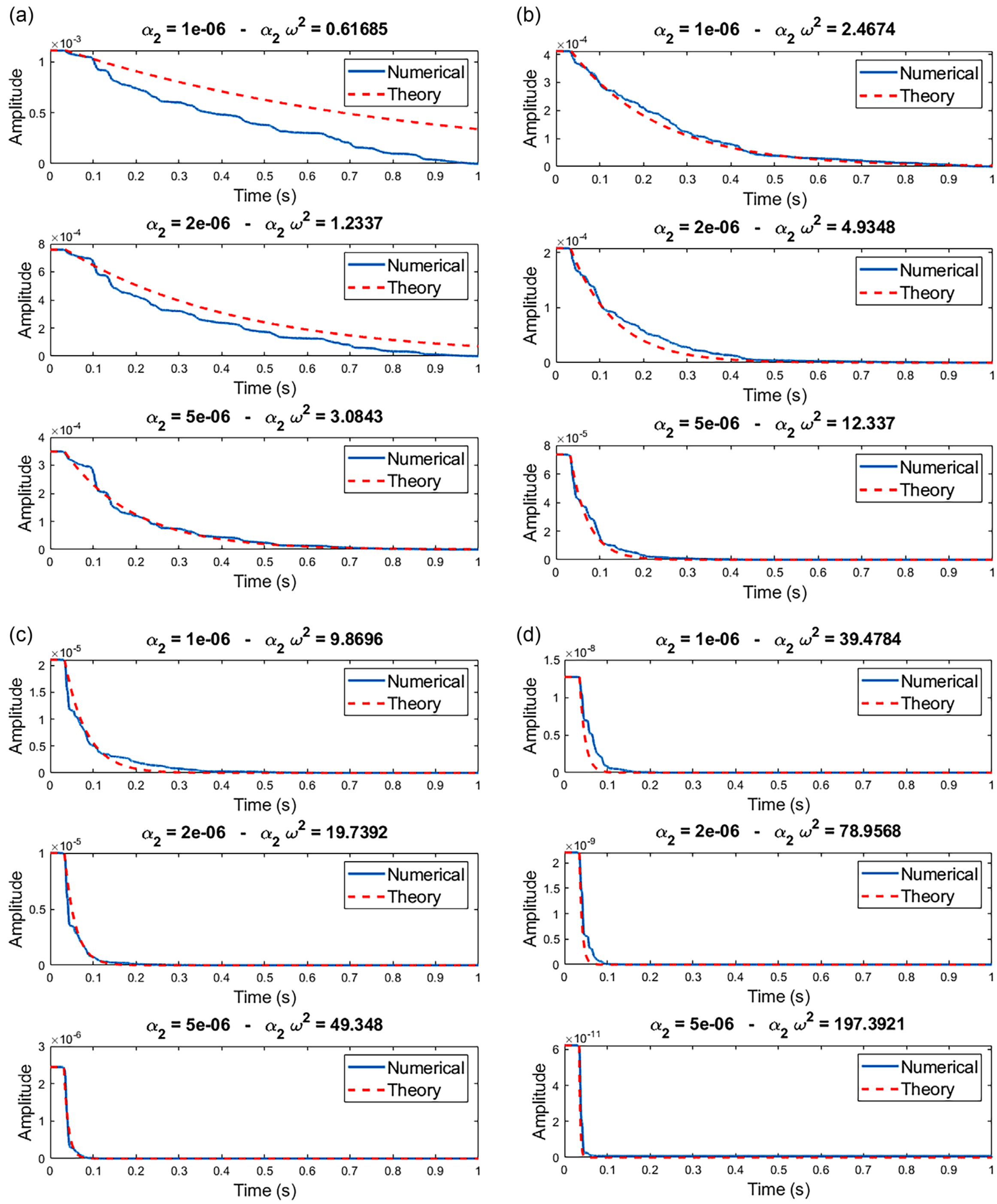

FIG. 6. (Color online) Comparison between the experimental energy decay curves with the theoretical energy decay curves in four octave bands.

presented in Raghuvanshi *et al.* (2009). Future work could focus on rigorous mathematical analysis of the scheme convergence and accuracy to ensure that the improved ARD method consistently provides reliable and precise results across different scenarios.

Further improvements in accuracy may involve developing an improved interface handling scheme that introduces less dispersion and reduces spurious reflections. Moreover, ARD could be embedded into a hybrid simulation technique to improve efficiency.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: IMPLEMENTATION OF PML

To model partial absorption of the acoustic pressure on the boundaries of the computational domain, we consider in the sequel PML conditions. In practice, we solve problem (2) in a domain $\hat{\Omega} = \Omega \cup \Omega_{\mathrm{PML}}$, where $\hat{\Omega} = [0, \ell_x'] \times [0, \ell_y'] \times [0, \ell_z']$ with $\ell_w' > \ell_w$ for $w = x, y, z$ and $\Omega_{\mathrm{PML}} = \hat{\Omega} \backslash \Omega$. Following Grote and Sim (2010), we consider the modified wave equation

$$\begin{cases} \dfrac{\partial v}{\partial t}(\underline{x}, t) - \mathrm{tr}(\Gamma_1) v(\underline{x}, t) + \mathrm{tr}(\Gamma_3) p(\underline{x}, t) - c^2 \Delta p(\underline{x}, t) \\ \quad = \nabla \cdot \phi\, \underline{x}(\underline{x}, t) - \zeta_x \zeta_y \zeta_z \psi(\underline{x}, t), \quad t \in \mathbb{R}^+, \underline{x} \in \hat{\Omega}, \\ \dfrac{\partial p}{\partial t}(\underline{x}, t) = v(\underline{x}, t), \qquad t \in \mathbb{R}^+, \underline{x} \in \hat{\Omega}, \\ \dfrac{\partial \underline{\phi}}{\partial t}(\underline{x}, t) = \Gamma_1 \underline{\phi}(\underline{x}, t) + c^2 \Gamma_2 \nabla p(\underline{x}, t) + c^2 \Gamma_3 \nabla \psi(\underline{x}, t), \\ \qquad t \in \mathbb{R}^+, \underline{x} \in \hat{\Omega}, \\ \dfrac{\partial \psi}{\partial t}(\underline{x}, t) = p(\underline{x}, t), \qquad t \in \mathbb{R}^+, \underline{x} \in \hat{\Omega}, \end{cases} \tag{A1}$$

where $\mathrm{tr}(\cdot)$ is the trace operator, $\underline{\phi}$ and ψ are auxiliary variables, and Γ_1, Γ_2, and Γ_3 are defined as

$$\Gamma_1 = \begin{bmatrix} -\zeta_x & 0 & 0 \\ 0 & -\zeta_y & 0 \\ 0 & 0 & -\zeta_z \end{bmatrix}, \quad \Gamma_3 = \begin{bmatrix} \zeta_y \zeta_z & 0 & 0 \\ 0 & \zeta_x \zeta_z & 0 \\ 0 & 0 & \zeta_x \zeta_y \end{bmatrix},$$

$$\Gamma_2 = \begin{bmatrix} \zeta_y + \zeta_z - \zeta_x & 0 & 0 \\ 0 & \zeta_x + \zeta_z - \zeta_y & 0 \\ 0 & 0 & \zeta_x + \zeta_y - \zeta_z \end{bmatrix}.$$

The functions $\zeta_w(w)$ for $w = \{x, y, z\}$ are positive inside the absorbing layer Ω_{PML} but vanish inside Ω. This damping profile can be chosen arbitrarily as a constant, linear, or quadratic function. An example of such a profile for $w = \{x, y, z\}$ is given by

$$\zeta_w(w) = \begin{cases} 0, \quad 0 \le w \le \ell_w, \\ \bar{\zeta}_w \left(\dfrac{w - \ell_w}{\ell_w' - \ell_w} - \dfrac{1}{2\pi} \sin\left(2\pi \dfrac{w - \ell_w}{\ell_w' - \ell_w} \right) \right), \\ \quad \ell_w < w < \ell_w', \\ \bar{\zeta}_w \left(\dfrac{w}{\ell_w - \ell_w'} - \dfrac{1}{2\pi} \sin\left(2\pi \dfrac{w}{\ell_w - \ell_w'} \right) \right), \\ \quad \ell_w - \ell_w' < w < 0, \end{cases}$$

where $\bar{\zeta}_w$ is a constant related to the relative reflection $R = [c/(\ell_w' - \ell_w)] \log(1/R)$, a measure of the amount of reflection that occurs when a wave encounters an interface. To approximate Eq. (A1), we proceed as follows. For any $(i, j, k) \in D_x \times D_y \times D_z$ and any $n \in \mathbb{N}$, we define

$$\varphi_{i,j,k}^{\hat{n}} = \frac{\varphi_{i,j,k}^{n+1/2} + \varphi_{i,j,k}^{n-1/2}}{2}, \quad \text{with} \quad \varphi_{i,j,k}^{n+1/2} = \frac{\varphi_{i,j,k}^{n+1} + \varphi_{i,j,k}^{n}}{2}$$

for $\varphi = \{\psi, p\}$, and set

$$(\nabla \cdot \underline{\phi})_{i,j,k}^n = \frac{1}{dh} \sum_{h=-2}^{2} b_h (\phi_{i+h,j,k}^n + b_h \phi_{i,j+h,k}^n + b_h \varphi_{i,j,k+h}^n),$$

where $b_{-2} = 1/12$, $b_{-1} = -2/3$, $b_0 = 0$, $b_1 = 2/3$, $b_2 = -1/12$. System (A1) is discretized by means of the following scheme:

$$v_{i,j,k}^{n+1} = \frac{\left(1 - \dfrac{dt}{2}(\zeta_{x;i} + \zeta_{y;j} + \zeta_{z;k})\right) v_{i,j,k}^n + dt c^2 (\Delta p)_{i,j,k}^n}{1 + \dfrac{dt}{2}(\zeta_{x;i} + \zeta_{y;j} + \zeta_{z;k})} - dt \frac{(\zeta_{y;j}\zeta_{z;k} + \zeta_{x;i}\zeta_{z;k} + \zeta_{x;i}\zeta_{y;j}) p_{i,j,k}^n - (\nabla \cdot \underline{\phi})_{i,j,k}^n + (\zeta_{x;i}\zeta_{y;j}\zeta_{z;k}) \psi_{i,j,k}^{\hat{n}}}{1 + \dfrac{dt}{2}(\zeta_{x;i} + \zeta_{y;j} + \zeta_{z;k})}$$

for the first equation, and

$$p_{i,j,k}^{n+1} = p_{i,j,k}^n + dt v_{i,j,k}^{n+1}, \quad \psi_{i,j,k}^{n+1/2} = \psi_{i,j,k}^{n-1/2} + dt p_{i,j,k}^n,$$

$$\phi_{i,j,k}^{n+1} = \frac{\left(1 - \dfrac{dt}{2}\zeta_{x;i}\right) \phi_{i,j,k}^n + dt(\zeta_{y;j} + \zeta_{z;k} - \zeta_{x;i}) \dfrac{p_{i,j,k}^{n+1/2} - p_{i,j,k}^{n+1/2}}{dh} + dt(\zeta_{y;j}\zeta_{z;k}) \dfrac{\psi_{i,j,k}^{n+1/2} - \psi_{i,j,k}^{n+1/2}}{dh}}{1 + \dfrac{dt}{2}\zeta_{x;i}}$$

for the remaining equations.

[1]See https://github.com/jinnsjj/ARD-simulator (Last viewed May 28, 2024).
[2]See https://github.com/mazamin7/WASAbi (Last viewed May 28, 2024).

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